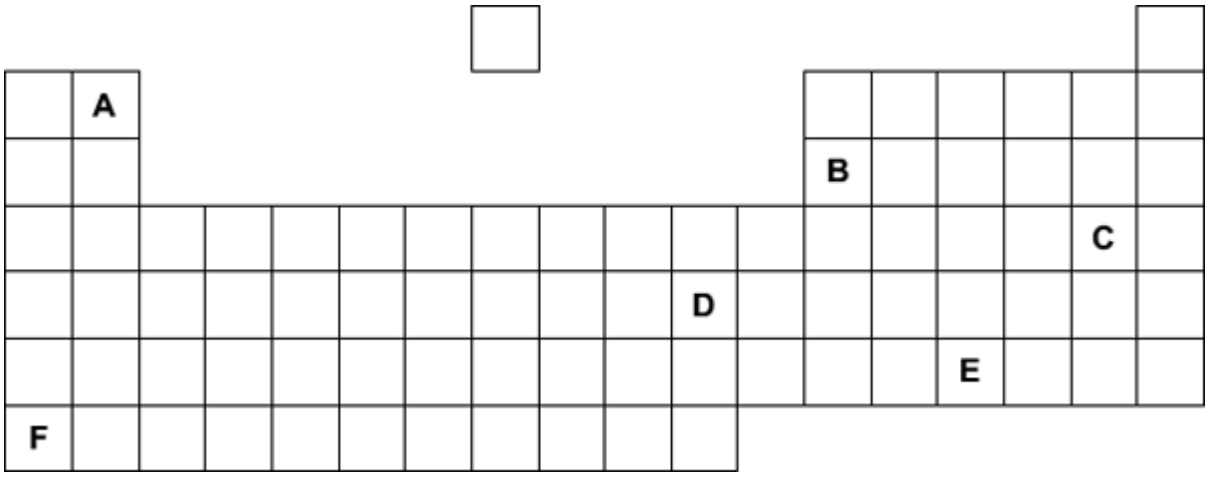
Q1.

The periodic table on the Data Sheet may help you to answer these questions.

(a) Part of the periodic table is shown below.



The letters are **not** the symbols of the elements.

Choose your answers **only** from the letters shown in the periodic table above.

Which letter, **A**, **B**, **C**, **D**, **E** or **F**, represents

(i) aluminium ☐

(1)

(ii) a Group 5 element ☐

(1)

(iii) an alkali metal ☐

(1)

(iv) the element with atomic (proton) number of 47 ☐

(1)

(v) an element with seven electrons in its outer shell? ☐

(1)

(b) The table shows the boiling points of the Group 7 elements.

The elements are arranged **in alphabetical order**.

Group 7 element		
Name	**Symbol**	**Boiling point in °C**
Astatine	At	337
Bromine		58
Chlorine	Cl	-34
Fluorine	F	-188
Iodine	I	184

(i) The symbol for bromine is missing from the table.

What is the symbol for bromine? Symbol = _____

(1)

(ii) Arrange these elements in order of **decreasing** boiling point. The first one and the last one have been done for you.

At _____ _____ _____ F

Highest boiling point → Lowest boiling point

(1)

(c) The table shows some statements about Group 7 elements.

Tick (✓) the **two** correct statements.

	Tick (✓)
They are halogens.	
They are metals.	
They become less reactive down Group 7.	
They are compounds.	

(2)

(Total 9 marks)

Q1.

(a) (i) B

1

(ii) E

1

(iii) F

1

(iv) D

1

(v) C

1

(b) (i) Br

do not accept BR or br or bR
ignore numbers
allow written in table if answer blank

1

(ii) I Br Cl

allow iodine, bromine, chlorine
allow I,B,C
allow capitals or lower case
allow 184, 58, -34
ignore numbers

1

(c) they are halogens

1

they become less reactive down Group 7

1

[9]